Bach Phan-Tat

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Computational researcher working on interpretable, theory-driven modelling of change, with a primary focus on semantic and conceptual change. My work combines computational linguistics, statistical modelling, research software, and methodological evaluation, with developing interests in complex systems and time series.

RESEARCH & PROFESSIONAL EXPERIENCE

KU Leuven, QLVL Research Group **Sep 2024 – Present**
Doctoral Researcher, Computational Linguistics *Leuven, Belgium*
Project: Underlying Dimensions in Conceptual Change

- Developed SynFlow, an open-source Python toolkit for multidimensional lexical semantic change analysis using structured linguistic patterns, distribution comparison, and incremental clustering of static embeddings.
- Built automated pipelines for LLM-assisted linguistic annotation and for cleaning, lemmatising, and parsing large corpora.
- Optimised computational workflows through parallel and batch processing, reducing processing time by 80%.
- Maintained NephoSem, a lexical-semantics toolkit developed by the QLVL research unit, and added support for other types of static embedding.

The Digital Humanities Institute, University of Sheffield **Sep 2026 – Nov 2026**
Visiting Research Scholar *Sheffield, England*
Project: Linguistic DNA

- Refined statistical techniques for ranking mutual information and frequency.
- Utilised AWS EMR to execute and monitor distributed data processing pipelines, parsing massive historical corpora for diachronic analysis.
- Developed graph visualisation and clustering techniques to support the exploration and interpretation of conceptual change.

Murf AI (speech technology company) **Nov 2024 – Nov 2026**
Remote Research Consultant *Utah, USA*
Project: Multilingual Text-to-Speech system

- Developed phoneme inventories and pronunciation lexicons for Vietnamese, Turkish, and four English varieties (US, UK, Scottish, and Australian), using corpus analysis and phonological rules to support multilingual TTS development.
- Designed evaluation pipelines and structured test sets for sentence segmentation models, including linguistic error categorisation and systematic analysis of model failures.
- Built Python workflows for text normalisation, lexical analysis, pronunciation-data preparation, and automated quality checks.
- Conducted error analysis across linguistic resources and model outputs, identifying recurring issues and translating findings into data-, lexicon-, and model-level recommendations.
- Collaborated with other teams on annotation, transcription, and data preparation for speech and language workflows.

Vbee (speech technology and conversational AI company) **Feb 2023 – Aug 2024**
Junior Research Engineer *Hanoi, Vietnam*
Project: HUST Chatbot (a conversational AI for Hanoi University of Science and Technology)

- Developed and evaluated machine-learning and BERT-based models for chat domain classification, toxicity detection, and name-based classification, achieving over 85% accuracy on held-out evaluation sets.
- Designed data-sampling and human-evaluation procedures, applying statistical methods to assess dataset quality, annotation reliability, and model performance.
- Built an LLM-assisted chatbot evaluation pipeline, including evaluation criteria, prompt design, automated response scoring, and systematic analysis of model outputs.
- Developed language-data pipelines including web crawling, information extraction, LLM-assisted data generation, preprocessing, annotation, and quality control for text and speech datasets.

- Contributed to the development of linguistic resources, and rule-based components for speech and language tasks, including speech-to-text, text-to-speech, and part-of-speech parsing.

what3words (location-mapping technology company) **Jul 2023 – Aug 2023**
Remote Research Consultant *London, England*
Project: Vietnamese Automatic Speech Recognition system (in collaboration with VinFast)
- Conducted systematic linguistic error analysis of Vietnamese ASR outputs, categorising recurring errors involving phonetic confusions, lexical ambiguity, and contextual misrecognition.
- Translated error-analysis findings into data- and model-level recommendations.

University of Stirling **Sep 2021 – Sep 2022**
Master Student *Stirling, Scotland*
Project: The structure of thời-gian
- Crawled, cleaned, and compiled a specialised corpus of 500 randomly sampled Vietnamese news articles.
- Identified seven senses of the Vietnamese lexeme *thời gian* ('time'), their semantic structure and semantic development using the Principled Polysemy framework.

EDUCATION

KU Leuven **Sep 2024 – Present**
PhD, Computational Linguistics *Leuven, Belgium*
Thesis: **Underlying Dimensions in Conceptual Change**
Thesis supervisors: Prof. Dr. Dirk Speelman, Prof. Dr. Dirk Geeraerts, Dr. Kris Heylen

WorldQuant University **Oct 2025 – Present**
MSc, Financial Engineering *Online*
Average grade (current): 84%

University of Stirling **Sep 2021 – Sep 2022**
MSc, English Language and Linguistics *Stirling, Scotland*
Thesis: **The structure of *thời gian*: A semantic analysis of the Vietnamese lexeme *thời gian* based on Principled Polysemy**
Thesis supervisor: Dr. Andrew Smith
Thesis grade: 80% (Distinction; class rank: 1st)
Average grade: 78% (Distinction; class rank: 1st)

Academy of Finance **Aug 2016 – Aug 2020**
BA, Corporate Finance *Hanoi, Vietnam*
Thesis grade: 9.2/10
Average grade: 7.15/10

PUBLICATIONS

Methods

- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, and Dirk Speelman. 2026. ReFRAME or Remain: Unsupervised Lexical Semantic Change Detection with Frame Semantics. *Proceedings of the 15th Joint Conference on Lexical and Computational Semantics (*SEM 2026), ACL 2026.*
- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, and Dirk Speelman. 2026. Transparent Semantic Change Detection with Dependency-Based Profiles. *Proceedings of the 6th International Workshop on Computational Approaches to Historical Language Change (LChange'26), EACL 2026.*

Software & toolkits

- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, Dirk Speelman. 2026. SynFlow: A Multidimensional Diachronic Semantic Analysis Toolkit. (Preprint).

Datasets, benchmarks & resources

- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, Dirk Speelman. 2026. Evaluating the Evaluator: Problems with SemEval-2020 Task 1 for Lexical Semantic Change Detection. *Journal of Open Humanities*

Data.

- Khai Le-Duc, Tuyen Tran, **Bach Phan-Tat**, et al. 2025. MultiMed-ST: Large-scale Many-to-many Multilingual Medical Speech Translation. *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP 2025)*.
- Khai Le-Duc, Phuc Phan, Tan-Hanh Pham, **Bach Phan-Tat**, Minh-Huong Ngo, Thanh Nguyen-Tang, Truong-Son Hy. 2025. MultiMed: Multilingual Medical Speech Recognition via Attention Encoder Decoder. *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics, Industry Track (ACL 2025)*. Top 15% of accepted papers.

Applications

- Catherine Wong, **Bach Phan-Tat**, Susan Fitzmaurice. 2026. Methods, Data, and Conceptual Change: Reflections from Two Quantitative Diachronic Case Studies. *Journal of Open Humanities Data*.
- Sofía Aguilar-Valdez[*], **Bach Phan-Tat**[*], Dirk Speelman, Dirk Geeraerts, and Stefania Degaetano-Ortlieb. 2026. Discursive parallels of the chemical revolution: topic modelling and distributional analysis. In *Understanding Science with Large Language Models? Potentials for the History, Philosophy, and Sociology of Science* (open access, transcript). [*]Equal contribution.
- Khai Le-Duc, Khai-Nguyen Nguyen, **Bach Phan-Tat**, Le Duy, Long Vo-Dang, Truong-Son Hy. 2025. Sentiment Reasoning for Healthcare. *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics, Industry Track (ACL 2025)*. Oral presentation.
- Duc Cao-Dinh, Khai Le-Duc, Anh Dao, **Bach Phan-Tat**, Chris Ngo, Duy M. H. Nguyen, Nguyen X. Khanh, Thanh Nguyen-Tang. 2025. Audio-3DVG: Unified Audio–Point Cloud Fusion for 3D Visual Grounding. (Preprint).

SELECTED PRESENTATIONS

- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, and Dirk Speelman. 2026. Underlying Dimensions in Semantic Change. Digital Humanities Congress 2026.
- **Bach Phan-Tat**, Dirk Geeraerts, and Dirk Speelman. 2026. Conceptual Change during the Chemical Revolution: Air, Acid, and Water in the Royal Society Corpus. ICAME 47.
- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, and Dirk Speelman. 2026. From Parsers to Prompts: Combining AI and Linguistics for Interpretable Language Evolution. AI in Language Evolution Workshop, Evolang 2026.
- Khai Le-Duc, Khai-Nguyen Nguyen, **Bach Phan-Tat**, et al. 2024. Sentiment Reasoning for Healthcare. Advancements In Medical Foundation Models: Explainability, Robustness, Security, and Beyond Workshop, NeurIPS 2024.
- **Bach Phan-Tat**. 2026. Underlying Dimensions in Semantic Change. NLP & text mining seminar organised by Leuven.AI.

ACADEMIC SERVICE & MENTORING

Supervising & mentoring

- Advised Master's theses, including TESOL at Nottingham Trent University (2026, Distinction) and Linguistics at Lancaster University (2024, Distinction).
- Mentored applicants in postgraduate and fellowship applications, with subsequent admissions to the Linguistics PhD programme at Lancaster University (2024), the YSEALI Fellowship (2024), the Stipendium Hungaricum Scholarship (2026).

Reviewing

- Reviewer for Cognitive Linguistic Studies (2026), Linguists' Day, Linguistic Society of Belgium (2025).

Organising

- Co-organised the PhD Day for the Department of Linguistics at KU Leuven. 2026.

AWARDS & HONOURS

- **2026: First Place, Databricks Challenge** (US$1,000), Hack-Nation's 4th Global AI Hackathon; 3,000+ participants from 300+ universities across 65+ countries.
- **2024: Marie Skłodowska–Curie Actions Doctoral Fellowship** (fully funded), Horizon Europe; awarded to 10 out of 284 applicants worldwide.

- **2022: Research-Based Learning Prize for the best Master's dissertation in Literature and Languages** (GBP 225), University of Stirling; the first recipient in the history of the Linguistics department.
- **2021: Postgraduate Scholarship** (GBP 4,000), University of Stirling.

COMPUTATIONAL & RESEARCH SKILLS

Programming & computing

- Python (NumPy, Pandas, Matplotlib, scikit-learn, spaCy, PyTorch, Hugging Face), R, Bash, SQL, MongoDB, Git/GitHub, LaTeX
- Parallel, batch, and GPU computing
- Cloud computing: Amazon Web Services (AWS), Google Cloud Platform (GCP)
- Containerisation: Docker

Computational methods

- Exploratory data analysis, data preprocessing and (interactive) data visualisation
- Machine learning and deep learning for NLP, structured data, time-series and computer vision
- Quantitative research design, experimental design, and A/B testing
- LLM-assisted annotation and evaluation workflows

Linguistics & corpus methods

- Corpus tools: SynFlow, CQPweb, #LancsBox, Sketch Engine, AntConc
- Linguistic analysis: phonetics, morphology, syntax, semantics
- Linguistic annotation, transcription, and corpus preparation

SELECTED TRAINING

Data science

- Applied Data Science Lab (WorldQuant University)
- Applied AI Lab: Deep Learning for Computer Vision (WorldQuant University)
- NeuroAI (Neuromatch Academy)

Neuroscience

- Computational Neuroscience (Neuromatch Academy)

Philosophy

- Philosophy and the Sciences: Introduction to the Philosophy of Cognitive Sciences (The University of Edinburgh)
- Philosophy and the Sciences: Introduction to the Philosophy of Physical Sciences (The University of Edinburgh)

LANGUAGES

- Vietnamese: Native
- English: IELTS 8.5 – C2
- Dutch: A2